

Fintech Onboarding & VKYC Workflow Optimization Research

Overview

The ultimate objective of this research is to understand the difference between the traditional banks and fintech-based platforms in their customer onboarding process for VKYC. The traditional banks have successfully digitized the onboarding process, but the entire process still remains to be operationally heavy. Whereas on the other hand, fintech-based platforms they focus more on conversions rather than compliance. They have focused on the conversational and user experience of the people onboard.

The entire study particularly focuses on understanding:

- How onboarding workflows are structured,
- Where user friction emerges,
- How fintechs reduce perceived effort,
- How onboarding speed and simplicity can be improved without compromising compliance and security requirements.

The broader aim was to explore where the fintech way of onboarding could significantly improve the onboarding efficiency and whether it could reduce the user journey to mere 35-40 seconds before the actual VKYC call duration.

The following platforms that I chose to analyze were:

Traditional Banks

- HDFC Bank
- Yes Bank
- Kotak Mahindra Bank
- SBI (YONO)

Fintech Platforms

- Jupiter
- Fi

The research involved studying onboarding workflows, VKYC interaction patterns, onboarding transitions, interface behavior, autofill mechanisms, and the overall emotional and operational experience offered to users.

Understanding the Traditional Banking Workflow

Although the onboarding systems of traditional banks differed slightly in implementation, the overall onboarding structure across most platforms remained broadly similar.

A typical onboarding journey involved:

1. Personal detail filling
2. Aadhaar/PAN verification
3. OTP authentication
4. Multiple form confirmations
5. VKYC scheduling or VC link generation
6. Video KYC verification
7. Backend verification and activation

The research helped me to come to the fact that the overall user journey on traditional banking systems is heavily compliance first. There is a strong focus on security and verification systems, but from user perspective it feels operationally heavy. One of the major observations that I made was the majority of the onboarding time is consumed before the actual VKYC interaction. So, this “Pre-KYC” experience becomes the primary source of friction and abandonment.

HDFC Bank Analysis

Among the traditional banks analyzed, HDFC demonstrated one of the strongest backend verification structures, but also one of the heaviest onboarding experiences.

The VKYC process included:

- state and city verification,
- OTP verification,
- verification of the last four digits of the registered mobile number,
- application form confirmations,
- income verification,
- PAN showcase,
- image capture,
- signature verification,
- and final KYC validation.

The complete VKYC interaction itself took approximately **2 minutes and 30 seconds**.

However, the larger concern was not only the VKYC duration, but the onboarding effort required before the VC process even began. Users were required to move across multiple onboarding screens involving repetitive form filling and repeated verification requests. A noticeable pattern was the duplication of verification efforts. Certain information or OTP-based confirmations appeared multiple times across onboarding and VKYC stages, increasing perceived effort even if operationally necessary.

Key Observations

- Heavy form-filling workflow
- Multiple onboarding screens
- Repetitive authentication requests
- Strong verification systems but slower onboarding
- Significant pre-VKYC interaction load

The onboarding system prioritized compliance robustness, but at the cost of onboarding fluidity and speed perception.

YES Bank Analysis

YES Bank's onboarding journey attempted to present a simplified onboarding narrative to users through marketing and interface communication. The onboarding experience was framed as a simple three-step process:

- Personal Details
- Aadhaar
- Video Verification

However, the actual onboarding workflow complexity remained relatively similar to traditional banking systems. One of the more advanced implementations observed was the use of OCR (Optical Character Recognition) to extract user details during onboarding. Additionally, Yes Bank introduced a separate authentication layer immediately before the VKYC stage to validate previously entered onboarding information.

While this improved verification confidence, it also added another interaction checkpoint before the VC stage. A key insight from the research was the difference between "perceived onboarding simplicity" and "actual workflow complexity." Although the interface messaging simplified the process psychologically, the underlying onboarding structure still remained interaction-heavy.

Key Observations

- OCR-assisted detail extraction
- Separate pre-VKYC authentication layer
- Simplified onboarding communication
- Actual workflow complexity similar to traditional systems
- Significant onboarding effort still concentrated in pre-VKYC flow

Kotak Mahindra Bank Analysis

Kotak Mahindra Bank demonstrated comparatively more modern onboarding behavior among traditional banking systems. The platform adopted several mobile-first onboarding approaches and

incorporated autofill systems to reduce manual typing. Features such as Aadhaar autofill significantly improved onboarding continuity and reduced interaction effort. Kotak was also one of the earliest banks to normalize zero-balance onboarding experiences, making onboarding psychologically less restrictive for users.

My research identified some key inconsistencies in the interactions design. One of the major UX shortcoming was the FAQ experience, where each FAQ interaction brings the user to a new screen or page instead of having an expandable dropdown interaction. This created unnecessary navigation friction. Also, the users were required to complete the onboarding details and VKYC process in a single continuous session reducing the workflow flexibility.

Key Observations

- Mobile-first onboarding behavior
- Aadhaar autofill integration
- Reduced manual typing
- Better onboarding continuity
- FAQ interaction friction due to page redirections
- Limited workflow flexibility

Kotak's onboarding reflected early fintech-inspired design adoption, although certain interaction patterns still introduced avoidable friction.

SBI (YONO) Analysis

SBI's onboarding through YONO represented a large-scale attempt at digital transformation while still preserving traditional banking operational structures.

The onboarding flow broadly included:

1. Savings account opening
2. Mobile verification
3. PAN and Aadhaar verification
4. Personal detail filling
5. Branch selection
6. VKYC scheduling
7. VKYC completion

One of SBI's strongest advantages was the trust factor associated with the brand itself. Users generally demonstrated lower hesitation levels while sharing personal or verification-related information due to SBI's institutional familiarity.

However, the onboarding workflow itself remained operationally traditional in nature. The research identified several friction points like multiple onboarding screens, repeated confirmations, branch selection steps, queue waiting uncertainty, onboarding interruptions, and inconsistent technical smoothness.

Branch selection was identified as an unnecessary manual interaction point since nearest branch allocation could potentially be automated. Another major issue was the lack of real-time queue transparency during VKYC waiting periods. Users were often left uncertain regarding expected waiting time, current queue position, or estimated agent availability. This uncertainty significantly increases abandonment probability.

Reviews and onboarding experiences also highlighted long waiting queues, onboarding screen freezes, app responsiveness issues, and onboarding interruptions. The estimated pre-VKYC effort itself ranged between approximately 4–8 minutes before queue waiting duration was even considered.

Key Observations

- Strong institutional trust factor
- Digitally enabled but traditionally structured onboarding
- Multiple onboarding screens and confirmations
- Weak queue transparency
- Queue abandonment risks
- Poor onboarding continuity during waiting periods
- Operational friction despite digitalization

A major insight from SBI's onboarding was that digitalization alone does not automatically create optimized workflows if the underlying process architecture remains unchanged.

Fintech Platform Analysis

Unlike traditional banks, fintech onboarding systems demonstrated a fundamentally different onboarding philosophy. Rather than treating onboarding purely as a compliance workflow, fintech platforms approached onboarding as a conversion-focused user journey.

The primary focus shifted towards minimizing perceived effort, reducing cognitive load, maintaining onboarding continuity, and emotionally reassuring users throughout the journey.

Jupiter Analysis

Jupiter's onboarding flow strongly emphasized conversational onboarding patterns and progressive interaction design. The onboarding experience followed a "one-question-per-screen" approach, significantly reducing visible complexity at any given stage.

The platform effectively utilized:

- progressive disclosure,
- inline validations,
- smart defaults,
- auto keyboard switching,
- bottom-sheet interactions,

- animated progress reassurance,
- and minimal upfront information collection.

Rather than overwhelming users with large forms, onboarding effort was distributed gradually across smaller interaction moments. Psychological reassurance mechanisms were also continuously present through progress indicators, simplified language, and guided transitions.

Key Observations

- Conversational UX
- One-question-per-screen approach
- Strong progressive disclosure
- Reduced visible effort
- Inline validations and smart defaults
- Smooth onboarding continuity
- Strong psychological reassurance

A major learning from Jupiter was that onboarding simplicity is often driven more by perceived effort reduction than actual workflow reduction.

Fi Analysis

Fi's onboarding workflow demonstrated one of the strongest examples of friction masking and speed perception optimization.

The platform heavily minimized text density and relied on:

- strong visual hierarchy,
- one-thumb usability,
- near-zero typing interactions,
- real-time PAN and Aadhaar fetching,
- and guided onboarding transitions.

The onboarding system continuously reinforced speed and simplicity psychologically through interaction pacing and interface behavior.

Fi also invested significantly in emotional reassurance through:

- conversational tone,
- simplified interaction language,
- and early trust-building patterns.

The estimated pre-VKYC interaction time was approximately 45–90 seconds, substantially lower than traditional onboarding systems.

Key Observations

- Minimal text-heavy interactions
- Very low typing effort
- Real-time data fetching
- Strong visual hierarchy
- Friction masking techniques
- Emotional reassurance focus
- Fast onboarding perception

Fi's onboarding highlighted how onboarding speed perception can often matter as much as actual onboarding duration.

Major Friction Areas Identified

Across the onboarding systems analyzed, the following friction points appeared consistently:

- repetitive form filling,
- excessive manual typing,
- repeated OTP/authentication requests,
- weak onboarding continuity,
- abrupt VKYC transitions,
- onboarding uncertainty,
- poor queue visibility,
- interaction-heavy verification flows,
- onboarding abandonment risks,
- and high cognitive load.

The research revealed that many onboarding drop-offs are not caused by a single major issue, but rather by the accumulation of multiple small friction points throughout the onboarding journey.

Proposed Product Philosophy

Based on the research findings, the proposed onboarding approach was designed around a fintech-first onboarding philosophy focused on reducing visible friction while maintaining compliance integrity.

The proposed design philosophy emphasized:

- Mobile-first onboarding
- Reduced typing effort

- Progressive disclosure
- Inline validations
- Smart defaults
- One-thumb usability
- Queue transparency
- Psychological reassurance
- Background verification systems
- Conversion-focused onboarding continuity.

Rather than exposing compliance-heavy workflows directly to users, the system aimed to minimize visible operational complexity through smoother interaction design.

Key Learnings

This research highlighted that onboarding optimization is not merely a technical challenge, but also a behavioural, psychological, and operational design challenge.

Some of the major learnings included:

- compliance-heavy systems significantly increase cognitive load,
- onboarding continuity strongly impacts completion rates,
- fintech onboarding succeeds by minimizing perceived effort,
- psychological reassurance is critical during onboarding,
- queue transparency directly affects abandonment probability,
- and small friction points compound into major onboarding drop-offs over time.

The study ultimately reinforced the idea that effective onboarding systems are not defined only by security and compliance strength, but also by how effortlessly users are guided through the journey.